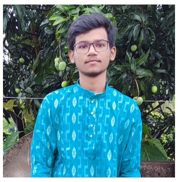

Mamun Chowdhury

UG (III Year II Semester)
Integrated M.Tech. Geophysical Technology
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Registration No: 23411024/2026

Indian Institute of Technology Roorkee

Area of Interest

Applied Machine Learning | Probability & Statistical Inference

Education

Year	Degree/Examination	Institution/Board	CGPA/ Percentage
2026	Int. M.Tech 3rd Year	Indian Institute of Technology, Roorkee	7.593
2023	Intermediate (Class XII)	JAWAHAR NAVODAYA VIDYALAYA, MURSHIDABAD	93.00 %
2021	Matriculate (Class X)	JAWAHAR NAVODAYA VIDYALAYA, MURSHIDABAD	95.80 %

Projects

Credit Card Behavior Score Prediction using Risk-Based Classification | IIT ROORKEE April 2025 - June 2025

- Developed a residual learning-based ML model (XGBoost over base predictions) to forecast road accident risk.
- Engineered spatial-temporal and contextual features from traffic and simulated data to improve predictive performance.
- Optimized model accuracy using error correction via residual boosting over baseline models.
- Performed EDA and feature importance analysis to identify key factors influencing accident risk.
- Applied probabilistic adjustments to handle prediction uncertainty.
- Generated insights on high-risk conditions and accident patterns to support proactive road safety planning.

Smart Product Pricing ML Challenge 2025 | Amazon ML Challenge 2025 October 2025

- Built a multimodal regression system to predict product prices using textual descriptions and product images (150K samples dataset).
- Extracted semantic features using text embeddings and TF-IDF, and visual representations using image embeddings.
- Designed a hybrid ensemble combining gradient boosting models with multimodal feature inputs.
- Implemented a stacking meta-model to aggregate predictions from multiple base learners and improve generalization.
- Performed EDA and feature analysis across text and image modalities to enhance predictive signal.
- Developed an end-to-end pipeline for preprocessing, feature fusion, model training, and evaluation.

Road Accident Risk Prediction using XGBoost over Residuals | Kaggle (Playground Series - Season 6 Episode 4) March 2026 - April 2026

- Developed an end-to-end machine learning pipeline to predict road accident risk using gradient boosting (XGBoost) with residual modeling for improved accuracy.
- Implemented a residual learning framework, where base model errors were modeled using XGBoost to enhance predictive performance over traditional approaches.
- Generated and utilized synthetic datasets to simulate real-world traffic and accident scenarios, enabling robust training and evaluation.
- Incorporated Bayesian correction techniques to refine predictions and handle uncertainty in accident risk estimation.
- Built a modular pipeline covering data preprocessing, feature engineering, model training, and evaluation.
- Tools & Tech: Python, XGBoost, Scikit-learn, Pandas, NumPy

Awards / Scholarships / Academic Achievements

- DAKSHANA Scholarship Programme 2023
- JNV Regional Chess 2019
- Silver Medel at Colors Trophy , IIT Roorkee (Hockey)

Skills

Computer languages	C++, Python
Software Packages	PyTorch, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib
Additional Courses	Introduction to Cognitive Science
Languages Known	Hindi (Native), Bengali (Native), English (Professional Working Proficiency)

Positions of Responsibility & Extra Curriculars

Executive Member | Himalayan Explorers' Club, IIT Roorkee November 2024 - Present

- Fosters adventure and exploration, organizing treks, expeditions, and outdoor activities, inspiring students to discover nature, build endurance, and develop leadership skills.

Member | NSO (Hockey), IIT Roorkee August 2023 - Present

- Promotes physical fitness and sportsmanship among students by offering structured training in various sports, fostering discipline, teamwork, and a healthy lifestyle.

Junior Associate | THOMSO, IIT Roorkee September 2023 - December 2023

- Embodies the spirit of creativity, cultural diversity, and youthful exuberance, making it a hallmark event in the Indian collegiate festival landscape.